

OKA MANIFOLDS ARE UBIQUITOUS

SICHENG AN, BIN GUO, PENG-CHAO WANG, AND SONG-YAN XIE

ABSTRACT. We prove that every smooth complex projective rationally connected manifold, and hence every smooth complex Fano manifold, is Oka by applying the analytic criteria of Du, Guo, Wang, and Xie. We also establish the Oka property for the smooth Calabi–Yau threefolds in Schoen’s construction: fiber products over the projective line of relatively minimal rational elliptic surfaces with sections and disjoint sets of singular values. For a smooth hypersurface of degree d in $\mathbb{P}^n$, its complement is Oka, equivalently holomorphically elliptic, if and only if $d \leq n + 1$. A connected smooth projective variety with reduced simple normal crossings boundary has Oka complement if it admits a nonconstant rational curve whose inverse image of the boundary consists of at most one point and whose pulled-back logarithmic tangent bundle is ample. The main positive results follow from holomorphic families of entire curves constructed by deformations of rational curves, holomorphic actions along genus-one fibers, and successive polar equations. The obstruction in higher degree is due to Carlson and Griffiths.

1. INTRODUCTION AND MAIN RESULTS

We establish the Oka property for rationally connected projective manifolds, smooth Schoen Calabi–Yau threefolds, and several classes of divisor complements. In particular, we determine exactly which smooth hypersurfaces in projective space have Oka complement. These results rest on three geometric constructions: deformations of rational curves, holomorphic actions along genus-one fibers, and lines with prescribed orders of contact with a hypersurface. The analytic criteria of Du, Guo, Wang, and Xie [1] turn the resulting holomorphic families of entire curves into the Oka property; we state them in the form needed here in Section 2.

Oka theory concerns existence, approximation, and interpolation for holomorphic maps from Stein spaces; see [10]. We use Forstnerič’s characterization by the convex approximation property (CAP) [9, Theorem 0.1], recalled in Section 2.

1.1. Compact projective manifolds. A connected smooth complex projective manifold X is *rationally connected* if some Zariski open dense subset $U \subset X \times X$ has the following property: for every $(x_1, x_2) \in U$ there is a morphism $u : \mathbb{P}^1 \rightarrow X$ whose image contains both x_1 and x_2 . A *Fano manifold* is a connected smooth complex projective manifold whose anticanonical line bundle $K_X^{-1} = \det TX$ is ample. For these definitions see [11, §§1–2]. Campana [14] and Kollár–Miyaoka–Mori [15] proved that smooth complex Fano manifolds are rationally connected.

Campana and Winkelmann [6, Theorem A] constructed entire curves that are dense in the usual complex topology and whose images contain any prescribed countable subset of a rationally connected projective manifold. Alarcón and Forstnerič [7, Definition 1.1] introduced the Oka-1 property, which concerns approximation and interpolation for maps from open Riemann surfaces, and conjectured that every rationally connected projective manifold has it [7, Conjecture 9.1]. Du, Guo, and Xie [8, Corollary 1.8] proved this conjecture. The following theorem establishes the full Oka property, for maps from Stein spaces of arbitrary finite dimension.

2020 *Mathematics Subject Classification.* 32Q56, 32E30, 14M22, 14J45.

Key words and phrases. Oka manifolds, rational connectedness, Calabi–Yau threefolds, hypersurface complements, logarithmic rational curves.

Theorem 1.1. *Every smooth complex projective rationally connected manifold is Oka. In particular, every smooth complex Fano manifold is Oka.*

The proof is given in Section 3, where deformations of very free rational curves give the families required by Theorem 2.2 after restriction to $\mathbb{C} \subset \mathbb{P}^1$.

Our second class consists of the smooth fiber products of rational elliptic surfaces in Schoen's construction [20]. We use the term *Schoen threefold* here for the fiber products satisfying the hypotheses below. Their Calabi–Yau properties in (i) are classical; our contribution is the entire map and the Oka conclusion in (ii). We recall the geometric argument for (i) in Section 4.

Theorem 1.2. *Let $p_i : S_i \rightarrow \mathbb{P}^1$, $i = 1, 2$, be relatively minimal elliptic fibrations with connected fibers and sections on smooth projective rational surfaces. Let $\Delta_i \subset \mathbb{P}^1$ be the set of singular values of p_i , and suppose that $\Delta_1 \cap \Delta_2 = \emptyset$. Then*

$$X = S_1 \times_{\mathbb{P}^1} S_2$$

has the following properties.

- (i) *It is a smooth connected projective simply connected threefold with $K_X \simeq \mathcal{O}_X$ and $H^1(X, \mathcal{O}_X) = H^2(X, \mathcal{O}_X) = 0$.*
- (ii) *There are a holomorphic map $F : \mathbb{C}^3 \rightarrow X$ and a dense Zariski open subset $U \subset X$ such that $U \subset F(\mathbb{C}^3)$ and dF_z is an isomorphism for every $z \in F^{-1}(U)$. Moreover, X is Oka.*

Theorem 1.2 extends the compact examples beyond rationally connected manifolds. The map in (ii) covers a dense Zariski open subset and is locally biholomorphic at every preimage of that subset. Section 4 constructs it by composing holomorphic actions along the genus-one fibers.

1.2. Divisor complements. For smooth hypersurfaces in projective space, we obtain an exact degree criterion for both the Oka property and holomorphic ellipticity. The latter means the existence of a dominating holomorphic spray; we recall the definition in Section 2.

Theorem 1.3. *Let $D \subset \mathbb{P}^n$ be a smooth hypersurface of degree $d \geq 1$, where $n \geq 1$. The following conditions are equivalent:*

- (i) $\mathbb{P}^n \setminus D$ is Oka;
- (ii) $\mathbb{P}^n \setminus D$ is holomorphically elliptic;
- (iii) $d \leq n + 1$.

The proof is given in Section 6. For the positive direction, successive polar equations give local holomorphic families of lines meeting D in at most two distinct points. We then construct entire curves in the complements of D in these lines. For the negative direction we use the differential-degeneracy theorem of Carlson–Griffiths [25, Corollary (6.11)]. The positive examples include $\mathbb{P}^3$ minus a smooth quartic surface and $\mathbb{P}^4$ minus a smooth quintic threefold.

We also obtain a criterion for divisor complements in smooth projective varieties. A reduced divisor $D = \sum_i D_i$ with irreducible components D_i has *simple normal crossings* if each D_i is smooth and D is a union of coordinate hyperplanes in suitable local holomorphic coordinates. The local holomorphic sections of the logarithmic tangent bundle $T_X(-\log D)$ are the vector fields that preserve the ideal sheaf of each D_i .

Theorem 1.4. *Let X be a connected smooth complex projective variety, and let $D \subset X$ be a reduced simple normal crossings divisor. Suppose that a nonconstant morphism $f : \mathbb{P}^1 \rightarrow X$ satisfies*

$$f^{-1}(D) \subseteq \{\infty\}, \quad f^*T_X(-\log D) \text{ is ample.}$$

Then $X \setminus D$ is Oka.

Section 5 proves Theorem 1.4 by deforming rational curves while preserving the pullback divisors f^*D_i .

The existence results of Chen–Zhu [16] give geometric applications of this criterion. Theorem 5.3 shows that $X \setminus D$ is Oka whenever X is a connected smooth complex projective variety of dimension at least two and D is a smooth irreducible ample rationally connected divisor; this covers in particular the case where D is smooth, irreducible and ample and $-(K_X + D)$ is ample. Corollary 5.4 gives the range $1 \leq d \leq n$ for smooth hypersurfaces in $\mathbb{P}^n$. The polar construction of Section 6 also reaches $d = n + 1$, where the logarithmic tangent bundle has trivial determinant and its pullback cannot be ample. The same corollary applies when $X \subset \mathbb{P}^N$ is a smooth complete intersection of multidegree $(d_1, \dots, d_c)$ and dimension at least two and D is a smooth hypersurface section of degree $e \geq 1$ with $d_1 + \dots + d_c + e \leq N$.

For a reduced simple normal crossings divisor D on a connected smooth projective variety X , we call (X, D) *log Fano* if $-(K_X + D)$ is ample. Corollary 5.5 gives an Oka complement for such a pair with $D = D_1 + \dots + D_r$, $r \geq 1$, if every D_i is ample, their classes are linearly independent in $N^1(X)_{\mathbb{Q}}$, and $\dim X - r \geq 2$. Here $N^1(X)_{\mathbb{Q}}$ denotes the tensor product with $\mathbb{Q}$ of the group of Cartier divisor classes modulo numerical equivalence.

Section 7 treats further examples that rely on classification, on holomorphic fiber bundles, and on known results about sprays and toric varieties. Theorem 7.1 covers all smooth projective log Fano surfaces with nonempty reduced simple normal crossings boundary, and we also treat complements of a smooth quadric and a transverse hyperplane, together with complements of unions of torus-invariant prime divisors in smooth toric varieties.

1.3. Methods and related work. The analytic criterion behind our results uses holomorphic maps $h_j : V \times \mathbb{C} \rightarrow Y$, where $V \subset Y$ is open, such that $h_j(y, 0) = y$ and the vectors $\partial_t h_j(y, 0)$ span $T_y Y$. Theorem 2.2 derives the Oka property from these families, and Corollary 2.5 allows them to be constructed only outside a proper closed complex analytic subset. The geometric constructions occupy Sections 3–6, while Section 2 supplies the implication from entire curves to convex approximation in every finite source dimension.

Complex Lie groups and complex homogeneous spaces are basic examples of Oka manifolds. Gromov’s theorem on dominating holomorphic sprays [5] provides another general source. Section 7 uses these results, together with Oka properties of fiber bundles and toric varieties, to give further examples.

An and Xie study the Oka property for smooth Fano threefolds and rationally connected threefolds in [2, 3]. The scalar-substitution argument of [2, Lemma 4.3] is adapted in [1, Lemma 5.1], a key input to the criterion of Section 2. Wang, Xie, and Zhao [4] study related Oka properties of compact complex surfaces.

Xie [17, Theorem 1.1] proves ellipticity for cubic hypersurface complements in $\mathbb{P}^n$, $n \geq 2$, except when the reduced support is the union of three distinct hyperplanes containing a common codimension-two linear subspace; this includes all smooth cubic complements. Our Theorem 1.3 treats smooth hypersurfaces in every degree, and reaches the anticanonical degree $n + 1$ by a construction with successive polars.

2. ENTIRE CURVES AND THE OKA PROPERTY

The geometric arguments in Sections 3–6 use the analytic results of Du, Guo, Wang, and Xie [1]. We state the required local condition and deduce its consequences from that source. All complex manifolds are finite-dimensional, Hausdorff, and second countable. We may argue component by component, the zero-dimensional case being immediate. An entire curve in Y is a holomorphic map $\mathbb{C} \rightarrow Y$.

A complex manifold Y has the *convex approximation property* (CAP) if, for every integer $m \geq 1$, every compact convex set $K \subset \mathbb{C}^m$, every holomorphic map f from a neighborhood of

K to Y , and every $\epsilon > 0$, there is a holomorphic map $F : \mathbb{C}^m \rightarrow Y$ with

$$\sup_K d_Y(F, f) < \epsilon.$$

Here d_Y is the distance induced by a fixed complete Hermitian metric on Y . This property characterizes Oka manifolds [9, Theorem 0.1].

A *holomorphic spray* on Y is a holomorphic map $s : E \rightarrow Y$ from a holomorphic vector bundle $E \rightarrow Y$ such that $s(0_y) = y$ for every $y \in Y$. It is *dominating* if $d(s|_{E_y})_{0_y} : E_y \rightarrow T_y Y$ is surjective for every y . A manifold admitting such a spray is *holomorphically elliptic*, or simply *elliptic*. Gromov's Oka principle [5] implies that every holomorphically elliptic manifold is Oka.

We now describe the families of entire curves used in this paper.

Definition 2.1. Consider the following condition on an n -dimensional complex manifold Y . For every $x \in Y$, there are an open neighborhood $V \ni x$, an integer $N \geq 1$, and holomorphic maps

$$h_j : V \times \mathbb{C} \longrightarrow Y \quad (1 \leq j \leq N), \quad h_j(y, 0) = y,$$

such that the vectors

$$\partial_t h_j(y, 0) := d(t \mapsto h_j(y, t))_0 \left(\frac{\partial}{\partial t} \Big|_0 \right), \quad 1 \leq j \leq N,$$

span $T_y Y$ over $\mathbb{C}$ for every $y \in V$. Each h_j is defined on the full product $V \times \mathbb{C}$; its image need not lie in V .

The next criterion follows from [1, Proposition 5.2 and Theorem 2.2].

Theorem 2.2. *A complex manifold satisfying the condition in Definition 2.1 has the convex approximation property in every finite source dimension and is therefore Oka.*

Recall the local approximation property L_n of [1, Definition 2.1], where $n = \dim Y$. For a compact set K , the notation $\mathcal{O}(K, Y)$ denotes the maps to Y that are holomorphic on some open neighborhood of K . For $r > 0$ write $\Delta_r = \{t \in \mathbb{C} : |t| < r\}$. The notation $A \Subset U$ means that $\bar{A}$ is compact and contained in U .

Definition 2.3. We consider pairs $D \subset D' \subset \mathbb{C}$ of closed Jordan discs with piecewise smooth boundaries such that $D' = D \cup E$ for another closed Jordan disc E with piecewise smooth boundary, where $D \cap E$ is a proper closed arc in both ∂D and ∂E .

The manifold Y has property L_n at x if there is an open neighborhood $W \ni x$ such that, for every such pair $D \subset D'$, every integer $q \geq 0$, every prescribed pair of closed polydiscs $P_0 \Subset \text{int } P \subset \mathbb{C}^q$, every $f \in \mathcal{O}(D \times P, W)$, and every $\epsilon > 0$, there is an $F \in \mathcal{O}(D' \times P_0, Y)$ satisfying

$$\sup_{(z,p) \in D \times P_0} d_Y(F(z, p), f(z, p)) < \epsilon.$$

For $q = 0$ the parameter polydiscs reduce to a point.

The subscript n records the target dimension; the parameter dimension q is arbitrary. The neighborhood W is independent of the disc pair, of the parameter polydiscs, of the map, and of the approximation error. The inner polydisc P_0 is prescribed before the approximant is chosen. The approximant takes values in Y and need not remain in W .

Proposition 2.4. *If Y satisfies the condition in Definition 2.1, then Y has property L_n at every point.*

Proof. Let $n = \dim Y \geq 1$ and $x \in Y$. Among the families of Definition 2.1, select n whose derivatives at $(x, 0)$ form a basis of $T_x Y$, and relabel them $h_1, \dots, h_n$. On the same fixed neighborhood V , their restrictions

$$h_{j,\nu} = h_j|_{V \times \Delta_\nu} : V \times \Delta_\nu \longrightarrow Y, \quad j = 1, \dots, n, \quad \nu \geq 1,$$

satisfy $h_{j,\nu}(y, 0) = y$, agree with h_j on $V \times \Delta_1$, and have scalar radii $\nu \rightarrow \infty$. Thus [1, Proposition 5.2] gives L_n at x , for every q and every prescribed inner polydisc P_0 . $\square$

Proof of Theorem 2.2. Proposition 2.4 gives L_n at every point. The local-to-global theorem [1, Theorem 2.2] yields CAP in every finite source dimension, and [9, Theorem 0.1] then gives the Oka property. $\square$

The criterion also allows a proper analytic exceptional set, by the removability result of [1, Corollary 5.4].

Corollary 2.5. *Suppose the families of Definition 2.1, with target Y , exist at every point outside a closed complex analytic subset $A \subset Y$ that is proper in each connected component. Then Y is Oka.*

Proof. The proof of Proposition 2.4 gives L_n at every point of $Y \setminus A$, with approximants taking values in Y ; its argument is local, so we may work on a connected component. On such a component the germ of A is proper at every point. Thus the points where Y fails to have L_n are locally contained in a proper analytic subset. Corollary 5.4 of [1] gives L_n everywhere, and Theorem 2.2 of [1] gives CAP in every finite source dimension, hence the Oka property. $\square$

In Corollary 2.5 the maps h_j take values in Y and may meet A ; the corollary therefore does not require $Y \setminus A$ itself to be Oka. The extension from an Oka open complement is [1, Theorem 1.5].

3. RATIONALLY CONNECTED AND FANO MANIFOLDS

Deformations of very free rational curves give the families required by Theorem 2.2 at every point of a rationally connected projective manifold, and these families are holomorphic on the whole product $V \times \mathbb{P}^1$.

A morphism $u : \mathbb{P}^1 \rightarrow X$ is *very free* if u^*TX is ample. By the Birkhoff–Grothendieck splitting theorem this means that

$$u^*TX \simeq \bigoplus_{\nu=1}^{\dim X} \mathcal{O}_{\mathbb{P}^1}(a_\nu), \quad a_\nu \geq 1;$$

see [13, Definition 3.1]. Here $\mathcal{O}_{\mathbb{P}^1}(a)$ denotes the line bundle of degree a .

Proposition 3.1. *Every smooth complex projective rationally connected manifold X satisfies the condition in Definition 2.1. More precisely, for every $x \in X$ the maps in that definition can be chosen to be holomorphic maps $h_j : V \times \mathbb{P}^1 \rightarrow X$, restricted afterwards to $V \times \mathbb{C}$.*

Proof. Let $n = \dim X \geq 1$ and $x \in X$, and let $e_1, \dots, e_n$ be a basis of $T_x X$, which specifies the tangent directions to be realized. Write t for the standard affine coordinate on $\mathbb{P}^1 \setminus \{\infty\} = \mathbb{C}$. The theorem on very free rational curves with prescribed tangent directions [11, Theorem 2.2(ii)] applies to X . Its single-point case, applied to the lines $\mathbb{C}e_j$ separately, gives morphisms

$$u_j : \mathbb{P}^1 \longrightarrow X, \quad u_j(0) = x, \quad (du_j)_0 \left(\frac{\partial}{\partial t} \Big|_0 \right) \in \mathbb{C}^* e_j, \quad u_j^*TX \text{ ample}.$$

Here a source reparametrization sends the prescribed point to 0. Write $u_j^*TX \simeq \bigoplus_{\nu=1}^n \mathcal{O}_{\mathbb{P}^1}(a_{j\nu})$ with $a_{j\nu} \geq 1$. Thus

$$H^1(\mathbb{P}^1, u_j^*TX) = 0, \quad H^1\left(\mathbb{P}^1, u_j^*TX \otimes \mathcal{O}_{\mathbb{P}^1}(-[0])\right) = 0,$$

since the corresponding line-bundle summands have degrees $a_{j\nu}$ and $a_{j\nu} - 1$, respectively. The notation $[0]$ denotes the divisor consisting of the point $0 \in \mathbb{P}^1$.

Let $\mathcal{H}$ be a finite-type open subscheme of $\text{Hom}(\mathbb{P}^1, X)$ containing the finitely many points $[u_j]$, and write $\mathcal{H}^{\text{an}}$ for its complex analytification. The source $\mathbb{P}^1$ is fixed. By the deformation

theory of morphisms [12, Proposition 1.6 and Theorem 1.9], $\mathcal{H}$ is smooth at $[u_j]$, with tangent space $H^0(\mathbb{P}^1, u_j^*TX)$. Evaluation at 0 gives $\text{ev}_0 : \mathcal{H}^{\text{an}} \rightarrow X$, $[u] \mapsto u(0)$, whose differential at $[u_j]$ is

$$d(\text{ev}_0)_{[u_j]} : H^0(\mathbb{P}^1, u_j^*TX) \longrightarrow T_x X, \quad s \longmapsto s(0).$$

It is surjective by the displayed vanishing after twisting by $-[0]$. Therefore ev_0 is a holomorphic submersion on an analytic neighborhood of $[u_j]$. The holomorphic implicit function theorem now gives holomorphic local sections of ev_0 on an analytic neighborhood of x ; shrinking these neighborhoods, we obtain sections

$$\sigma_j : V \longrightarrow \mathcal{H}^{\text{an}}, \quad \text{ev}_0 \circ \sigma_j = \text{id}_V, \quad \sigma_j(x) = [u_j].$$

Composing the sections with the universal evaluation morphism [12, §1.4] gives holomorphic maps

$$h_j : V \times \mathbb{P}^1 \longrightarrow X, \quad h_j(y, t) = \sigma_j(y)(t), \quad h_j(y, 0) = y.$$

Their derivatives at $(x, 0)$ lie in the lines $\mathbb{C}e_j$ and are nonzero. By continuity, after shrinking V we have

$$\frac{\partial h_1}{\partial t}(y, 0) \wedge \cdots \wedge \frac{\partial h_n}{\partial t}(y, 0) \neq 0 \quad (y \in V).$$

Consequently these vectors span $T_y X$ for every $y \in V$. Restricting the second variable to $\mathbb{C} = \mathbb{P}^1 \setminus \{\infty\}$ gives the maps required in Definition 2.1. $\square$

Proof of Theorem 1.1. Proposition 3.1 and Theorem 2.2 give the Oka property for every smooth complex projective rationally connected manifold. Every smooth complex Fano manifold is rationally connected by Kollár–Miyazaki–Mori [15, Theorem 0.1], which gives the Fano case. $\square$

4. SMOOTH SCHOEN CALABI–YAU THREEFOLDS

We consider the smooth fiber products satisfying the hypotheses of Theorem 1.2. Their Calabi–Yau geometry comes from Schoen’s construction [20]; we recall the needed facts in part (i) of the proof. To prove the Oka property, we construct a holomorphic map from $\mathbb{C}^3$ whose image contains a dense Zariski open subset and which is locally biholomorphic at every preimage of that subset, then apply Corollary 2.5. The two genus-one fibrations give holomorphic actions on the fibers by contraction with a volume form, as in Xie–Zhao [22, Lemma 3.1 and Proposition 3.3].

The following lemma constructs the actions over both regular and critical values of the fibrations.

Lemma 4.1. *Let $f : Y \rightarrow B$ be a proper holomorphic map from a smooth complex threefold with a nowhere-vanishing holomorphic three-form Ω to a smooth complex surface, and write $\text{Tot}(K_B)$ for the total space of the canonical line bundle of B . There is a holomorphic map*

$$\mathcal{A}_f : Y \times_B \text{Tot}(K_B) \longrightarrow Y$$

whose restriction over each $b \in B$ is an action of the additive group $K_{B,b}$ on $f^{-1}(b)$. Its infinitesimal vector is determined by

$$\iota_{V(y,\eta)} \Omega_y = (df_y)^* \eta, \quad \eta \in K_{B,f(y)}.$$

If $f^{-1}(b)$ is nonempty and connected and df has rank two at every point of this fiber, then the action on $f^{-1}(b)$ is transitive. Under these hypotheses, for every $y \in f^{-1}(b)$ and every $0 \neq \eta \in K_{B,b}$, the orbit map $t \mapsto \mathcal{A}_f(y, t\eta)$ has nonzero derivative at every $t \in \mathbb{C}$.

Proof. Contraction with Ω identifies TY with $\bigwedge^2 T^*Y$, so that V depends holomorphically and linearly on η . Where df has rank two, V is vertical and is nonzero whenever $\eta \neq 0$; where the rank is at most one, $V = 0$. On the smooth total space $E = \text{Tot}(f^*K_B) \subset Y \times \text{Tot}(K_B)$, the vector field $W(y, \eta) = (V(y, \eta), 0)$ is therefore well defined and tangent to E . The projection

$E \rightarrow \text{Tot}(K_B)$, being a base change of f , is proper, and W is vertical for it. Every trajectory stays in a compact fiber, which prevents finite-time escape. The local holomorphic flows agree by uniqueness and give a jointly holomorphic flow for every complex time. The argument takes place on the smooth manifold E and therefore also applies over singular fibers.

The time-one map of this flow, followed by projection to Y , is $\mathcal{A}_f$. For fixed b , the parameters lie in the one-dimensional space $K_{B,b}$, so the corresponding vector fields on $f^{-1}(b)$ are scalar multiples of one field. Linearity in η and uniqueness of the flow therefore give

$$\begin{aligned} f(\mathcal{A}_f(y, \eta)) &= f(y), & \mathcal{A}_f(y, 0) &= y, \\ \mathcal{A}_f(\mathcal{A}_f(y, \eta), \eta') &= \mathcal{A}_f(y, \eta + \eta') \end{aligned}$$

for $\eta, \eta' \in K_{B,f(y)}$. On a connected fiber where df has rank two everywhere, the infinitesimal vector field associated with a nonzero η never vanishes, so each orbit map has nonzero derivative at every time and each orbit is open in the fiber. Connectedness of the fiber then makes the action transitive. $\square$

Proof of Theorem 1.2. (i) Write s_i for the sections, $f_i : X \rightarrow S_i$ for the projections, and $p = p_i \circ f_i$. At every point of X at least one of dp_1, dp_2 is surjective, so X is the transverse inverse image of the diagonal in $\mathbb{P}^1 \times \mathbb{P}^1$ and is smooth of dimension three. Since X is a closed subvariety of $M = S_1 \times S_2$ it is projective, and it is connected because p has connected fibers and connected base.

A section excludes multiple fibers. The canonical bundle formula for relatively minimal elliptic surfaces gives $K_{S_i} \simeq p_i^*(K_{\mathbb{P}^1} \otimes \mathcal{L}_i)$, where $\mathcal{L}_i = (R^1(p_i)_* \mathcal{O}_{S_i})^\vee$ has degree $\chi(\mathcal{O}_{S_i}) = 1$. Thus $K_{S_i} \simeq p_i^* \mathcal{O}_{\mathbb{P}^1}(-1)$; see also [21, Section 3.1, p. 506]. The diagonal has bidegree $(1, 1)$, so $\mathcal{O}_M(X) \simeq K_M^{-1}$. Adjunction gives $K_X \simeq \mathcal{O}_X$ and a nowhere-vanishing three-form Ω . Rationality and the Künneth formula give $H^j(M, \mathcal{O}_M) = 0$ for $j > 0$, and Serre duality gives $H^j(M, K_M) = 0$ for $j < 4$. The exact sequence $0 \rightarrow K_M \rightarrow \mathcal{O}_M \rightarrow \mathcal{O}_X \rightarrow 0$ therefore gives $H^1(X, \mathcal{O}_X) = H^2(X, \mathcal{O}_X) = 0$.

For simple connectedness we follow the argument of Gachet–Lin–Wang [23, Lemma 4.8]. Put $B^\circ = \mathbb{P}^1 \setminus (\Delta_1 \cup \Delta_2)$. The smooth proper fibration $p^{-1}(B^\circ) \rightarrow B^\circ$ has a section, namely (s_1, s_2) , and fiber $X_b = S_{1,b} \times S_{2,b}$. Its fundamental group is generated by the image of $\pi_1(X_b)$ under the fiber inclusion and the image of $\pi_1(B^\circ)$ under the section. The inclusion $p^{-1}(B^\circ) \hookrightarrow X$ induces a surjection on fundamental groups, since loops can be perturbed off the omitted complex divisor. The image of $\pi_1(B^\circ)$ in $\pi_1(X)$ under the section is trivial because the section extends over $\mathbb{P}^1$. The images of the two factors of $\pi_1(X_b)$ in $\pi_1(X)$ are trivial as well, since their inclusions into X factor through the simply connected rational surfaces via

$$x_1 \mapsto (x_1, s_2(p_1(x_1))), \quad x_2 \mapsto (s_1(p_2(x_2)), x_2).$$

Hence $\pi_1(X) = 1$.

(ii) Choose $b_\infty \in \mathbb{P}^1$ and a biholomorphism $\alpha : \mathbb{C} \rightarrow \mathbb{P}^1 \setminus \{b_\infty\}$. The product of the two sections gives

$$j(z) = (s_1(\alpha(z)), s_2(\alpha(z))) \in X.$$

Apply Lemma 4.1 to the proper projections f_1, f_2 , and denote the resulting actions by $\mathcal{A}_1, \mathcal{A}_2$. The canonical bundle formula above gives

$$(f_1 \circ j)^* K_{S_1} \simeq \alpha^* \mathcal{O}_{\mathbb{P}^1}(-1).$$

Since $\mathcal{O}_{\mathbb{P}^1}(-1)$ is trivial on $\mathbb{P}^1 \setminus \{b_\infty\}$, we may choose a nowhere-vanishing holomorphic section e_1 of $(f_1 \circ j)^* K_{S_1}$, giving the map

$$G : \mathbb{C}^2 \rightarrow X, \quad G(z, u) = \mathcal{A}_1(j(z), u e_1(z)).$$

Then $f_1(G(z, u)) = s_1(\alpha(z))$ and $p(G(z, u)) = \alpha(z)$. Writing $\text{pr}_z : \mathbb{C}^2 \rightarrow \mathbb{C}$ for the first projection, we likewise have

$$(f_2 \circ G)^* K_{S_2} \simeq \text{pr}_z^* \alpha^* \mathcal{O}_{\mathbb{P}^1}(-1),$$

so this line bundle is holomorphically trivial. Choose a nowhere-vanishing holomorphic section e_2 of this bundle and set

$$F : \mathbb{C}^3 \rightarrow X, \quad F(z, u, v) = \mathcal{A}_2(G(z, u), v e_2(z, u)).$$

Here $e_1(z) \in K_{S_1, f_1(j(z))}$ and $e_2(z, u) \in K_{S_2, f_2(G(z, u))}$, so these two applications of the actions give holomorphic maps on all of $\mathbb{C}^2$ and $\mathbb{C}^3$, respectively. Since the second action preserves f_2 , we also have $p(F(z, u, v)) = \alpha(z)$ for all $(z, u, v) \in \mathbb{C}^3$.

Let $U = p^{-1}(B^\circ \setminus \{b_\infty\})$. The sets Δ_i are finite and p is surjective, so U is a nonempty Zariski open subset of the smooth connected variety X , and hence is dense. We show that $U \subset F(\mathbb{C}^3)$ and that dF is invertible at every point of $F^{-1}(U)$. Given $(x_1, x_2) \in U$, choose z with $\alpha(z) = p(x_1, x_2)$. The first action is transitive on $\{s_1(\alpha(z))\} \times S_{2, \alpha(z)}$, so some u gives $G(z, u) = (s_1(\alpha(z)), x_2)$. The second action is transitive on $S_{1, \alpha(z)} \times \{x_2\}$, so some v gives $F(z, u, v) = (x_1, x_2)$.

Over B° , both projections f_i are submersions, being base changes of smooth elliptic fibrations. For the rank assertion put $H = f_2 \circ G$. Whenever $\alpha(z) \in B^\circ$, the vector $\partial_u H$ is nonzero and tangent to the elliptic fiber of p_2 , whereas $dp_2(\partial_z H) = d\alpha_z \neq 0$; thus dH is invertible there. The action identities give

$$f_2 \circ F(z, u, v) = H(z, u).$$

For $(z, u, v) \in F^{-1}(U)$ the vector $\partial_v F(z, u, v)$ is nonzero and spans $\ker d(f_2)_{F(z, u, v)}$. Applying df_2 to $dF(a, b, c) = 0$ gives $dH(a, b) = 0$, so $a = b = 0$, and nonvanishing of $\partial_v F$ then gives $c = 0$. Hence dF is invertible at every point of $F^{-1}(U)$.

At every $x \in U$, choose a local inverse branch $\ell : V \rightarrow \mathbb{C}^3$ of F , where $V \subset U$ is a neighborhood of x , and let $\varepsilon_1, \varepsilon_2, \varepsilon_3$ be the standard basis of $\mathbb{C}^3$. The holomorphic maps

$$h_j : V \times \mathbb{C} \rightarrow X, \quad h_j(y, t) = F(\ell(y) + t\varepsilon_j), \quad j = 1, 2, 3,$$

satisfy

$$h_j(y, 0) = y, \quad \partial_t h_j(y, 0) = dF_{\ell(y)}(\varepsilon_j).$$

These three derivatives form a basis of $T_y X$. Since $X \setminus U$ is a proper closed algebraic subset, Corollary 2.5 shows that X is Oka. $\square$

Such pairs of elliptic surfaces exist: a general automorphism of the base of one rational elliptic surface moves its finite set of singular values away from that of the other. These threefolds are not rationally connected. Indeed, a very free rational curve would have positive anticanonical degree, contrary to $K_X \simeq \mathcal{O}_X$.

5. LOGARITHMIC RATIONAL CURVES AND DIVISOR COMPLEMENTS

5.1. A criterion using logarithmic rational curves. We prove Theorem 1.4 by deforming rational curves while keeping the pullback divisors fixed, so that the restrictions of the curves to $\mathbb{C}$ avoid the boundary. The logarithmic tangent bundle controls the values and the first derivatives of the deformed curves.

Throughout this section, X is a connected smooth complex projective variety, $D = \sum_{i=1}^r D_i$ is a reduced simple normal crossings divisor, $U = X \setminus D$, and $n = \dim X$. We allow $r = 0$, in which case $D = 0$. Fix distinct points $0, \infty \in \mathbb{P}^1$ and an affine coordinate t on $\mathbb{P}^1 \setminus \{\infty\}$ with $t(0) = 0$, and write $g'(0) = dg_0(\partial/\partial t|_0) \in T_{g(0)}X$. The notation $[0]$ and $[\infty]$ denotes the corresponding point divisors.

The sheaf of holomorphic sections of the logarithmic tangent bundle $T_X(-\log D)$ is the locally free sheaf of vector fields preserving the ideal sheaf of each D_i . In coordinates $x_1, \dots, x_n$ with $D = \{x_1 \cdots x_k = 0\}$ it has the frame

$$x_1 \frac{\partial}{\partial x_1}, \dots, x_k \frac{\partial}{\partial x_k}, \frac{\partial}{\partial x_{k+1}}, \dots, \frac{\partial}{\partial x_n}.$$

A morphism $f : \mathbb{P}^1 \rightarrow X$ satisfying the hypotheses of Theorem 1.4 is called a *very free logarithmic rational curve*. Note that it may happen that $f(\mathbb{P}^1) \cap D = \emptyset$.

In particular, U is Oka if it is $\mathbb{A}^1$ -connected in the sense of Chen–Zhu [16, Definition 1.2]. The latter means that there are an algebraic variety T and a morphism $a : T \times \mathbb{A}^1 \rightarrow U$ such that every map $a_s : \mathbb{A}^1 \rightarrow U$ is nonconstant and proper and the two-point evaluation map

$$T \times \mathbb{A}^1 \times \mathbb{A}^1 \longrightarrow U \times U, \quad (s, t_1, t_2) \longmapsto (a(s, t_1), a(s, t_2)),$$

is dominant. For a smooth projective compactification with simple normal crossings boundary over $\mathbb{C}$, this condition implies the existence of a curve as in Theorem 1.4, by [16, Proposition 1.13(2)]; separability is automatic in characteristic zero. This notion of $\mathbb{A}^1$ -connectedness concerns algebraic curves and is distinct from motivic $\mathbb{A}^1$ -connectedness.

The geometric applications below include complements of smooth ample rationally connected divisors (Theorem 5.3) and low-degree hypersurface sections of projective complete intersections (Corollary 5.4).

5.2. Prescribed pullback divisors and first jets. Fix f as in Theorem 1.4 and write $f^*D_i = m_i[\infty]$ with $m_i \geq 0$. We work with parametrized maps, keeping the source $\mathbb{P}^1$, the points $0, \infty$, and the coordinate t fixed. Fix a projective embedding of X and the degree of f with respect to it; the corresponding scheme of morphisms $\mathbb{P}^1 \rightarrow X$ is of finite type. Restrict it to the open-and-closed locus on which $\deg g^*\mathcal{O}_X(D_i) = m_i$ for all i , and let H be the open subset of maps whose image is contained in no D_i . Pulling back the divisors gives an algebraic morphism

$$c : H \longrightarrow \prod_{i=1}^r \mathrm{Sym}^{m_i}(\mathbb{P}^1), \quad c(g) = (g^*D_i)_i.$$

The pullback of D_i under the universal evaluation map is a relative effective Cartier divisor of degree m_i on $H \times \mathbb{P}^1$, hence determines a morphism to the symmetric power $\mathrm{Sym}^{m_i}(\mathbb{P}^1)$. We take $\mathrm{Sym}^0(\mathbb{P}^1)$ and the empty product to be a point. Let

$$M = c^{-1}((m_i[\infty])_i)$$

be the scheme-theoretic fiber. When all $m_i = 0$, we have $M = H$ and $g^*T_X(-\log D) \simeq g^*T_X$ for every $g \in M$; in particular, the case $D = \emptyset$ recovers the ordinary very free curve calculation. Every $g \in M$ satisfies $g(\mathbb{C}) \subset U$, so the universal evaluation map restricts to a morphism

$$(5.1) \quad \mathcal{F} : M \times \mathbb{C} \longrightarrow U.$$

Lemma 5.1. *Let $g \in M$ be a map for which $E_g = g^*T_X(-\log D)$ is ample. Then M is smooth at g , with tangent space $T_g M \simeq H^0(\mathbb{P}^1, E_g)$. The first-jet map*

$$J : M \longrightarrow TU, \quad g \longmapsto (g(0), g'(0)),$$

is a submersion at g . In particular, evaluation at 0 is a submersion at g . On the fiber of evaluation at 0 over $g(0)$, derivative evaluation $\tilde{g} \mapsto \tilde{g}'(0) \in T_{g(0)}U$ has surjective differential at g .

Proof. The inclusion of the logarithmic vector fields in all vector fields gives an exact sequence

$$(5.2) \quad 0 \longrightarrow E_g \longrightarrow g^*T_X \longrightarrow Q_g \longrightarrow 0, \quad Q_g = \bigoplus_{i:m_i>0} g^*\mathcal{O}_X(D_i)|_{m_i[\infty]}.$$

To check exactness, choose simple normal crossings coordinates near $g(\infty)$, in which the components met by g are given by $x_i = 0$. In a source coordinate z centered at infinity,

$$x_i \circ g = z^{m_i} u_i(z), \quad u_i(0) \neq 0.$$

The corresponding logarithmic tangent vector maps to $z^{m_i} u_i(z) \partial / \partial x_i$. Thus the map is injective and its cokernel is the stated torsion sheaf. Away from infinity it is an isomorphism.

Ampleness gives $E_g \cong \bigoplus_{\nu=1}^n \mathcal{O}_{\mathbb{P}^1}(a_\nu)$ with $a_\nu \geq 1$, and therefore

$$(5.3) \quad H^1(\mathbb{P}^1, E_g) = H^1(\mathbb{P}^1, E_g(-2[0])) = 0.$$

Since Q_g is a torsion sheaf, (5.2) also gives $H^1(\mathbb{P}^1, g^*T_X) = 0$. Deformation theory for morphisms from $\mathbb{P}^1$ therefore shows that H is smooth at g , with $T_g H = H^0(\mathbb{P}^1, g^*T_X)$; see [12, Proposition 1.6 and Theorem 1.9].

For $Z_i = m_i[\infty]$, the tangent space to the symmetric power has the intrinsic description

$$T_{Z_i} \text{Sym}^{m_i}(\mathbb{P}^1) \simeq H^0(Z_i, \mathcal{O}_{\mathbb{P}^1}(Z_i)|_{Z_i}) \simeq H^0(\mathbb{P}^1, g^*\mathcal{O}_X(D_i)|_{Z_i}),$$

where the second identification uses $g^*D_i = Z_i$; both sides are zero when $m_i = 0$. Under these identifications the differential of c is the quotient map

$$H^0(\mathbb{P}^1, g^*T_X) \longrightarrow H^0(\mathbb{P}^1, Q_g).$$

Locally it sends an infinitesimal variation of $x_i \circ g$ to its class modulo z^{m_i} , which is the infinitesimal displacement of the zero divisor of $x_i \circ g$. The vanishing $H^1(\mathbb{P}^1, E_g) = 0$ makes this differential surjective. Since the product of symmetric powers is smooth, c is smooth at g and its fiber M is smooth at g with the asserted tangent space.

The chosen coordinate t identifies the space of first jets at 0 of germs of maps to X with TX . At the jet $J(g)$, the tangent space $T_{J(g)}TU$ is naturally identified with $H^0(\mathbb{P}^1, g^*T_X|_{2[0]})$. Since the divisor conditions are supported at ∞ , we have $E_g|_{2[0]} \simeq g^*T_X|_{2[0]}$. Under this identification and $T_g M \simeq H^0(\mathbb{P}^1, E_g)$, the differential dJ_g is the restriction map on global sections in the exact sequence

$$0 \longrightarrow E_g(-2[0]) \longrightarrow E_g \longrightarrow E_g|_{2[0]} \longrightarrow 0.$$

It is surjective by (5.3), so $J : M \rightarrow TU$ is a submersion at g . Composing with the bundle projection $TU \rightarrow U$ gives the assertion about evaluation at 0. Restricting J to the fiber of evaluation at 0 over $g(0)$ gives the assertion about derivative evaluation. $\square$

Proposition 5.2. *Under the hypotheses of Theorem 1.4 the condition in Definition 2.1 holds at every point of a nonempty Zariski open subset of U .*

Proof. Let $M^+ \subset M$ be the locus where E_g is ample; it contains f . On $\mathbb{P}^1$, ampleness is equivalent to $H^1(\mathbb{P}^1, E_g(-2[0])) = 0$. Semicontinuity applied to the pullback of $T_X(-\log D)$ under the universal map therefore shows that M^+ is Zariski open. By Lemma 5.1 the morphism

$$\text{ev}_0 : M^+ \longrightarrow U$$

is smooth, so its image U^+ is a nonempty Zariski open subset of U .

We now pass to the analytification of M^+ . Fix $x \in U^+$ and $g \in M^+$ with $g(0) = x$. Derivative evaluation on $\text{ev}_0^{-1}(x)$ is a submersion near g , so its image contains a nonempty Euclidean open subset of $T_x U$, and such a subset contains n linearly independent vectors. We may therefore choose maps $g_1, \dots, g_n \in M^+$ with $g_j(0) = x$ whose derivatives at 0 form a basis of $T_x U$.

For each j the holomorphic submersion ev_0 has a local holomorphic section $\sigma_j : V_j \rightarrow M^+$ with $\sigma_j(x) = g_j$. On the common neighborhood $V = \bigcap_j V_j$ of x set

$$h_j(y, t) = \mathcal{F}(\sigma_j(y), t), \quad (y, t) \in V \times \mathbb{C}.$$

By (5.1) these maps are holomorphic on $V \times \mathbb{C}$ with values in U , and $h_j(y, 0) = y$. Their derivatives $\partial_t h_j(y, 0)$ form a basis at x , hence also on a smaller common neighborhood of x , so the maps satisfy Definition 2.1. $\square$

Proof of Theorem 1.4. The complement $U = X \setminus D$ is connected. Apply Corollary 2.5 with $Y = U$, using Proposition 5.2 and the proper closed algebraic subset $A = U \setminus U^+$; it follows that U is Oka. $\square$

5.3. Geometric applications. Chen–Zhu’s theorem gives $\mathbb{A}^1$ -connectedness when a smooth projective pair has a rationally connected fully free deepest stratum [16, Theorem 1.3]. In the applications below that stratum is $Z = D_1 \cap \cdots \cap D_r$. The condition *fully free* means that there are finitely many free rational curves $a_j : \mathbb{P}^1 \rightarrow Z$ such that

$$c_{ij} = \deg a_j^*(\mathcal{O}_X(D_i)|_Z) \geq 0, \quad \text{rank}_{\mathbb{C}}(c_{ij}) = r;$$

see [16, Remark 1.4]. Here a rational curve $a : \mathbb{P}^1 \rightarrow Z$ is free if $a^*T_Z \simeq \bigoplus_{\nu} \mathcal{O}_{\mathbb{P}^1}(b_{\nu})$ with $b_{\nu} \geq 0$ for every ν .

Theorem 5.3. *If $D \subset X$ is a smooth irreducible ample rationally connected divisor and $\dim X \geq 2$, then $X \setminus D$ is Oka. The conclusion holds in particular when D is smooth, irreducible and ample, $\dim X \geq 2$, and $-(K_X + D)$ is ample.*

Proof. The smooth rationally connected divisor D carries a very free rational curve $a : \mathbb{P}^1 \rightarrow D$. When $\dim D = 1$ we have $D \simeq \mathbb{P}^1$, and the identity map under this identification suffices. Since D is ample,

$$\deg a^*\mathcal{O}_D(D) > 0.$$

Thus the one-row degree matrix has rank one, and D is a fully free deepest stratum. Chen–Zhu’s Theorem 1.3 gives $\mathbb{A}^1$ -connectedness, and their Proposition 1.13(2) supplies a curve satisfying the hypotheses of Theorem 1.4, so $X \setminus D$ is Oka.

If $-(K_X + D)$ is ample, adjunction gives

$$-K_D = -(K_X + D)|_D \quad \text{ample.}$$

Consequently D is Fano, hence rationally connected [14, 15], and the first assertion applies. $\square$

Corollary 5.4. *Every smooth hypersurface $D \subset \mathbb{P}^n$ of degree $1 \leq d \leq n$ has Oka complement. More generally, let $X \subset \mathbb{P}^N$ be a smooth complete intersection of multidegree $(d_1, \dots, d_c)$ and dimension at least two. If D is a smooth hypersurface section of degree $e \geq 1$ and*

$$d_1 + \cdots + d_c + e \leq N,$$

then $X \setminus D$ is Oka.

Proof. Write H for the hyperplane class of the ambient projective space. For $\mathbb{P}^n$ the log anticanonical class is $(n+1-d)H$; for the complete-intersection pair it is

$$-(K_X + D) = (N+1-d_1-\cdots-d_c-e)H|_X.$$

The stated inequalities make these classes ample. When $\dim X \geq 2$, the connectedness theorem for ample effective divisors makes the smooth ample divisor D connected, and smoothness then gives irreducibility; Theorem 5.3 applies. For $n = 1$ the hypersurface is a single point of $\mathbb{P}^1$, whose complement is $\mathbb{C}$. $\square$

For several boundary components, we use their classes in $N^1(X)_{\mathbb{Q}}$, the space of Cartier divisor classes with rational coefficients modulo numerical equivalence.

Corollary 5.5. *Suppose $(X, D_1 + \cdots + D_r)$ is a reduced simple normal crossings log Fano pair with $r \geq 1$, each D_i is ample, the classes $[D_1], \dots, [D_r]$ are linearly independent in $N^1(X)_{\mathbb{Q}}$, and $\dim X - r \geq 2$. Then $X \setminus D$ is Oka.*

Proof. The complete intersection $Z = \bigcap_i D_i$ is smooth and has dimension at least two, and successive applications of ample-divisor connectedness show that it is connected. Adjunction gives $-K_Z = -(K_X + D)|_Z$, so Z is Fano and rationally connected. Here ampleness gives positive degrees on nonconstant free curves in Z , while numerical independence gives the full-rank condition: the restrictions of the D_i remain numerically independent, and very free curve classes span the space of numerical curve classes on Z , as in [16, Proposition 3.12]. Chen–Zhu’s Theorem 1.5 [16] gives $\mathbb{A}^1$ -connectedness, and [16, Proposition 1.13(2)] together with Theorem 1.4 shows that the complement is Oka. $\square$

For example, let $X = \mathbb{P}^2 \times \mathbb{P}^3$ and let D_1, D_2 be smooth transverse divisors of bidegrees $(1, 1)$ and $(1, 2)$. Writing H_1, H_2 for the pullbacks of the hyperplane classes from the two factors, we have

$$[D_1] = H_1 + H_2, \quad [D_2] = H_1 + 2H_2, \quad -K_X = 3H_1 + 4H_2.$$

Both divisors are ample, their coefficient matrix has determinant 1, and $D_1 \cap D_2$ has dimension three. Since

$$-(K_X + D_1 + D_2) = H_1 + H_2$$

is ample, Corollary 5.5 shows that $X \setminus (D_1 \cup D_2)$ is Oka.

The criterion does not account for every reduced simple normal crossings log Fano pair. For example, $\mathbb{P}^2$ minus two lines is $\mathbb{C} \times \mathbb{C}^*$, which is homogeneous and Oka, but not $\mathbb{A}^1$ -connected, since every algebraic morphism $\mathbb{A}^1 \rightarrow \mathbb{C}^*$ is constant; its $\mathbb{C}^*$ factor nevertheless admits the holomorphic family $(y, t) \mapsto ye^t$. In dimension one a reduced log Fano pair consists of $\mathbb{P}^1$ with at most one boundary point, so its complement is $\mathbb{P}^1$ or $\mathbb{C}$.

For a smooth hypersurface $D \subset \mathbb{P}^n$ of degree $n + 1$,

$$\det T_{\mathbb{P}^n}(-\log D) = \mathcal{O}_{\mathbb{P}^n}(-(K_{\mathbb{P}^n} + D)) \cong \mathcal{O}_{\mathbb{P}^n}.$$

For every morphism $f : \mathbb{P}^1 \rightarrow \mathbb{P}^n$, the bundle $f^*T_{\mathbb{P}^n}(-\log D)$ therefore has determinant of degree zero and cannot be ample. Thus Theorem 1.4 does not apply at this degree; Section 6 treats it by means of successive polar equations.

Positivity of $-(K_X + B)$ for a fractional boundary B does not by itself imply that $X \setminus \text{Supp } B$ is Oka. Indeed, the $\mathbb{Q}$ -divisor $B = \frac{1}{2}([0] + [1] + [\infty])$ on $\mathbb{P}^1$ satisfies $\deg(-(K_{\mathbb{P}^1} + B)) = \frac{1}{2} > 0$, whereas $\mathbb{P}^1 \setminus \text{Supp } B = \mathbb{P}^1 \setminus \{0, 1, \infty\}$ is hyperbolic.

6. SMOOTH HYPERSURFACE COMPLEMENTS

We prove Theorem 1.3, which determines the degrees for which the complement of a smooth hypersurface in projective space is Oka, or equivalently holomorphically elliptic.

Corollary 5.4 gives the positive conclusion for $d \leq n$. The construction below also reaches the anticanonical degree $d = n + 1$, and uses lines whose intersection multiplicity with D at a point of D is at least $d - 1$. It works for every smooth hypersurface in the stated degree range, and only the projection center is required to be general. The obstruction in high degree is due to Carlson–Griffiths [25, Corollary (6.11)].

Lemma 6.1. *Let F be a homogeneous polynomial of degree d on $\mathbb{C}^N$, where $N \geq d \geq 3$, and suppose that $X = \{F = 0\} \subset \mathbb{P}^{N-1}$ is smooth. There is a nonempty Zariski open subset $\Omega \subset \mathbb{P}^{N-1} \setminus X$ such that, for every $[a] \in \Omega$ and any nonzero representative $a = (a_1, \dots, a_N)$, the closed subscheme*

$$Z_a = \{F = D_a F = \dots = D_a^{d-2} F = 0\} \subset \mathbb{P}^{N-1}, \quad D_a = \sum_{i=1}^N a_i \frac{\partial}{\partial v_i},$$

is a smooth nonempty complete intersection of multidegree $(d, d-1, \dots, 2)$, dimension $N-d$, and degree $d!$, whose points span $\mathbb{P}^{N-1}$. Moreover, for the incidence scheme

$$\mathcal{I} = \{([a], [v]) \in \Omega \times X : D_a F(v) = \dots = D_a^{d-2} F(v) = 0\},$$

the projection $\pi : \mathcal{I} \rightarrow \Omega$ is smooth and surjective, of relative dimension $N-d$.

Proof. Fix a hyperplane $L \simeq \mathbb{P}^{N-2}$. Projection $p_a : X \rightarrow L$ from a point $[a] \notin X \cup L$ is finite: a positive-dimensional fiber would contain the entire line through $[a]$, contradicting $[a] \notin X$. The kernel of its differential has dimension at most one, since the ambient projection has one-dimensional fibers. Mather’s generic-projection theorem, in the formulation of Alzati–Ballico–Ottaviani [24, Theorem 1.1 and Section 2, step (2)], applies with ambient projective dimension $N-1$, codimension one, and center dimension zero. For each $1 \leq k \leq d-2 \leq N-2$,

it gives a nonempty Zariski open set of centers for which the jet extension of p_a is transverse to the corank-one Thom–Boardman stratum with symbol $(1, \dots, 1)$ of length k . Intersecting these finitely many open sets gives a nonempty Zariski open set Ω on which all these transversality conditions hold. The hypersurface X remains fixed.

We identify the corresponding subschemes with the polar equations. Near a common zero $[v]$, choose a linear form ℓ with $\ell(a) = 0$ and $\ell(v) \neq 0$. On the affine chart $\ell = 1$, take coordinates (z, u, t) , with $z \in \mathbb{C}^{N-3}$, so that translation by a is translation in t and the projection forgets t . The equation of X is $P(z, u, t) = 0$, where $\partial_t^j P = (D_a^j F)|_{\ell=1}$. Since $\partial_t P = 0$ at the point and X is smooth, we can choose u so that $\partial_u P \neq 0$ there and write $u = g(z, t)$. After shrinking the coordinate neighborhood,

$$A(z, t) = \partial_u P(z, g(z, t), t)$$

is nowhere zero. Set $q_j = \partial_t^j P(z, g(z, t), t)$ and $h_j = \partial_t^j g$. Repeated differentiation of $P(z, g(z, t), t) = 0$ gives

$$q_j + Ah_j \in (h_1, \dots, h_{j-1}), \quad (q_1, \dots, q_j) = (h_1, \dots, h_j).$$

These are identities of ideals of holomorphic functions on the graph. At a common zero, the differentials of the q_j and h_j are related by an invertible triangular matrix. For $(z, t) \mapsto (z, g(z, t))$, the kernel on $h_1 = 0$ is generated by $\partial/\partial t$. Its tangency to $h_1 = \dots = h_j = 0$ imposes the next equation $h_{j+1} = 0$. Thus the successive corank-one conditions are $h_1 = \dots = h_k = 0$, and jet transversality makes their differentials linearly independent. Hence the polar equations cut out a smooth subscheme of codimension $d - 2$ in X .

Let J be the ideal generated by $F, D_a F, \dots, D_a^{d-2} F$ in $S = \mathbb{C}[v_1, \dots, v_N]$. Its projective zero set is nonempty: otherwise $\sqrt{J}$ would be the irrelevant ideal of height N , contrary to the height theorem for an ideal generated by $d - 1 < N$ elements. The dimension calculation above shows that $\text{ht } J = d - 1$. Thus the generators form a regular sequence, and S/J is Cohen–Macaulay of dimension $N - d + 1 \geq 1$. A general linear form λ is a nonzerodivisor on S/J . If a homogeneous polynomial b lies in the saturation of J , then $\lambda^r b \in J$ for some r , whence $b \in J$; therefore J is saturated. The projective scheme is smooth, hence reduced. Since J is saturated, a linear form vanishing at all its points would belong to J ; but J has no element of degree one, because all its generators have degree at least two. Thus no hyperplane contains its points. Its degree as a complete intersection is $d(d - 1) \cdots 2 = d!$.

Each equation $D_a^j F(v) = 0$ is bihomogeneous of bidegree $(j, d - j)$ in (a, v) , so the incidence scheme is independent of the chosen projective representatives. The differentials with respect to $[v]$ are independent. The relative Jacobian criterion therefore makes $\pi : \mathcal{I} \rightarrow \Omega$ smooth of relative dimension $N - d$, and the nonemptiness of every fiber makes it surjective. $\square$

Proof of Theorem 1.3. Write $D = \{F = 0\}$ for a homogeneous polynomial F of degree d on $\mathbb{C}^{n+1}$. We treat degrees one and two separately below. Suppose first that $3 \leq d \leq n + 1$. Apply Lemma 6.1 with $N = n + 1$ and $X = D$, and write $y_0 = [a_0] \in \Omega$ with $a_0 \neq 0$. Since Z_{a_0} spans $\mathbb{P}^n$, we can choose points $[v_1^0], \dots, [v_n^0] \in Z_{a_0}$ whose representatives have classes forming a basis of $\mathbb{C}^{n+1}/\mathbb{C}a_0$. The smooth incidence projection has local holomorphic sections through the points $([a_0], [v_j^0]) \in \mathcal{I}$. Taking nonzero holomorphic lifts of their second components, together with a local holomorphic lift of the inclusion into $\mathbb{P}^n$, gives maps

$$a, v_1, \dots, v_n : V \longrightarrow \mathbb{C}^{n+1} \setminus \{0\}, \quad [a(y)] = y, \quad [v_j(y)] \in Z_{a(y)},$$

on a common coordinate neighborhood $V \subset \Omega$ of y_0 . After shrinking V , the classes of $v_1(y), \dots, v_n(y)$ modulo $\mathbb{C}a(y)$ form a basis for every $y \in V$.

For each fixed y , write $a = a(y)$ and $v = v_j(y)$. Homogeneity gives

$$[s^k]F(a + sv) = \frac{D_a^{d-k} F(v)}{(d - k)!} = 0 \quad (2 \leq k \leq d),$$

where $[s^k]$ denotes the coefficient of s^k . Consequently,

$$F(a + sv) = F(a) + s dF_a(v), \quad F(ra + sv) = r^{d-1}(rF(a) + s dF_a(v)).$$

Thus the line through $[a]$ and $[v]$ has intersection multiplicity at least $d - 1$ with D at $[v]$, and meets D in at most two distinct points.

The function

$$E(\lambda, t) = \begin{cases} \frac{e^{\lambda t} - 1}{\lambda}, & \lambda \neq 0, \\ t, & \lambda = 0, \end{cases}$$

is entire on $\mathbb{C}^2$ and satisfies $1 + \lambda E(\lambda, t) = e^{\lambda t}$. For $j = 1, \dots, n$ put

$$h_j(y, t) = [a(y) + F(a(y))E(dF_{a(y)}(v_j(y)), t)v_j(y)], \quad (y, t) \in V \times \mathbb{C}.$$

The value of F at the vector in brackets is

$$F(a(y)) \exp(dF_{a(y)}(v_j(y))t) \neq 0.$$

Hence $h_j : V \times \mathbb{C} \rightarrow \mathbb{P}^n \setminus D$ is well defined and holomorphic, with $h_j(y, 0) = y$. Its initial velocity is represented by $F(a(y))v_j(y)$ modulo $\mathbb{C}a(y)$, so these velocities form a basis of $T_y\mathbb{P}^n$. In this construction $a(y)$ and $v_j(y)$ are held fixed as t varies.

The complement $\mathbb{P}^n \setminus D$ is connected: the line joining two distinct points of the complement is not contained in D , hence meets D in a finite set, and the complement of that set in the line is connected. The construction works at every point of Ω , whose complement in $\mathbb{P}^n \setminus D$ is a proper closed algebraic subset. Corollary 2.5 therefore shows that $\mathbb{P}^n \setminus D$ is Oka.

For $d = 2$, the smooth quadric $\{F = 0\}$ spans $\mathbb{P}^n$. At any $y_0 \notin D$, choose a local holomorphic lift $a(y)$ of y and nonzero constant vectors $v_1, \dots, v_n$ with $F(v_j) = 0$ whose classes form a basis of $\mathbb{C}^{n+1}/\mathbb{C}a(y_0)$. After shrinking the neighborhood V of y_0 , use the same formula for h_j . The identity $F(a + sv) = F(a) + s dF_a(v)$ and the preceding argument apply unchanged. For $d = 1$, the complement is $\mathbb{C}^n$. These cases include $\mathbb{P}^1$ minus two points, which is $\mathbb{C}^*$.

If $d \geq n + 2$, then $\mathcal{O}_{\mathbb{P}^n}(K_{\mathbb{P}^n} + D) \cong \mathcal{O}_{\mathbb{P}^n}(d - n - 1)$ is ample. The smooth divisor $D \in |\mathcal{O}_{\mathbb{P}^n}(d)|$ therefore satisfies the hypotheses of Carlson–Griffiths [25, Section 6(a), Corollary (6.11)], with $V = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(d)$. Their corollary implies that every entire map $\mathbb{C}^n \rightarrow \mathbb{P}^n \setminus D$ has rank less than n everywhere. An Oka n -fold, on the other hand, admits an entire map from $\mathbb{C}^n$ whose differential has rank n at some point: approximate a local coordinate parametrization on a closed ball by CAP and apply Cauchy estimates on a smaller ball. Therefore $\mathbb{P}^n \setminus D$ is not Oka in this degree range.

Finally, D is an ample effective divisor, so $\mathbb{P}^n \setminus D$ is affine and hence Stein. On Stein manifolds, the Oka property and holomorphic ellipticity are equivalent by Gromov's theorem [5]; see [17, Section 2.2]. This gives the claimed equivalence with holomorphic ellipticity. $\square$

In particular, $\mathbb{P}^3$ minus any smooth quartic surface and $\mathbb{P}^4$ minus any smooth quintic threefold are holomorphically elliptic, and hence Oka. In the quintic case, Lemma 6.1 gives 120 distinct points spanning $\mathbb{P}^4$ for each projection center in a nonempty Zariski open set. These conclusions concern the complements of the compact hypersurfaces.

7. FURTHER EXAMPLES FROM BUNDLES, SPRAYS, AND TORIC VARIETIES

We conclude with complements whose Oka property follows from known results on holomorphic fiber bundles, dominating sprays, and smooth toric varieties. These examples also show that the ampleness hypothesis on the boundary in Theorem 5.3 and the numerical independence hypothesis in Corollary 5.5 are not necessary for a complement to be Oka.

7.1. Log Fano surfaces.

Theorem 7.1. *Let S be a smooth projective complex surface and let $D \neq 0$ be a reduced simple normal crossings divisor. If $-(K_S + D)$ is ample, then $S \setminus D$ is an Oka manifold.*

Proof. Write $\mathbb{F}_m$ for the Hirzebruch surface, C_m for a section with $C_m^2 = -m$, and F for a fiber of its ruling. Maeda's classification [18, §3.4, pp. 89–90] gives the following possibilities, up to isomorphism:

- (1) $S = \mathbb{P}^2$ and D is a line;
- (2) $S = \mathbb{P}^2$ and D is the union of two distinct lines;
- (3) $S = \mathbb{P}^2$ and D is a smooth conic;
- (4) $S = \mathbb{F}_m$, $m \geq 0$, and $D = C_m$;
- (5) $S = \mathbb{F}_m$, $m \geq 0$, and $D = C_m + F$;
- (6) $S = \mathbb{F}_1$ and D is a section with $D^2 = 1$;
- (7) $S = \mathbb{F}_0$ and D is a section with $D^2 = 2$.

The first two complements are $\mathbb{C}^2$ and $\mathbb{C} \times \mathbb{C}^*$, respectively. In the third case, identify $\mathbb{P}^2$ with the projectivization of the binary quadratic forms, so that the conic becomes the discriminant locus. Its complement parametrizes unordered pairs of distinct points of $\mathbb{P}^1$; the transitive action of $\mathrm{PGL}_2(\mathbb{C})$ makes it a complex homogeneous manifold, hence an Oka manifold.

In each Hirzebruch case the boundary consists of one section of the ruling and at most one fiber. The complement of a section in a holomorphic $\mathbb{P}^1$ -bundle is a holomorphic fiber bundle with fiber $\mathbb{C}$: locally on the base a holomorphic family of projective coordinate changes sends the section to infinity. If the boundary also contains a fiber, restrict the base to the complement of its image. Thus $S \setminus D$ is a holomorphic fiber bundle with fiber $\mathbb{C}$ over $\mathbb{P}^1$ or $\mathbb{C}$. Both base and fiber are Oka, so the total space is Oka by [9, Theorem 1.4]. $\square$

Remark 7.2. For the pairs $(\mathbb{F}_m, C_m)$ the boundary self-intersection is $-m$, which can be arbitrarily negative, so the ample-boundary assumption of Theorem 5.3 is not necessary for a complement to be Oka.

7.2. Quadric–hyperplane complements.

We use the spray terminology of Section 2.

The following special case of Xie's results [17, Theorem 6.1 and Corollary 6.3] illustrates how complete holomorphic vector fields produce a dominating spray, and we recall the explicit construction.

Proposition 7.3. *Let $n \geq 2$, let $Q \subset \mathbb{P}^n$ be a smooth quadric, and let $H \subset \mathbb{P}^n$ be a hyperplane transverse to Q . Then $\mathbb{P}^n \setminus (Q \cup H)$ is holomorphically elliptic, and hence Oka. For $n \geq 3$, the reduced simple normal crossings pair $(\mathbb{P}^n, Q + H)$ is log Fano, although its boundary classes are numerically dependent.*

Proof. In the affine chart $\mathbb{P}^n \setminus H$, the homogeneous quadratic part of the equation of Q defines $Q \cap H$. This intersection is smooth by transversality, so the quadratic part is nondegenerate. After completing the square, a zero constant term would make the origin a singular point of Q . A linear change of coordinates and rescaling therefore identify the complement with

$$U = \{z \in \mathbb{C}^n : q(z) \neq 1\}, \quad q(z) = z_1^2 + \cdots + z_n^2.$$

Let $B(z, v) = \sum_j z_j v_j$, so $q(z) = B(z, z)$. Choose a basis of isotropic vectors, that is, vectors satisfying $q(v_j) = 0$; for example,

$$v_1 = e_1 + ie_2, \quad v_2 = e_1 - ie_2, \quad v_j = e_1 + ie_j \quad (3 \leq j \leq n).$$

For each such v , set $a_v(z) = 2B(z, v)$ and consider the holomorphic vector field

$$V_v(z) = (q(z) - 1)v.$$

The identity $q(v) = 0$ gives

$$q(z + uv) - 1 = q(z) - 1 + u a_v(z),$$

and a_v is constant on $z + \mathbb{C}v$. Using the entire function E from the proof of Theorem 1.3, the flow of V_v is

$$\Phi_v(z, t) = z + E(a_v(z), t)(q(z) - 1)v, \quad (z, t) \in \mathbb{C}^n \times \mathbb{C}.$$

Indeed, the coefficient $u(t)$ of v satisfies $u'(t) = q(z) - 1 + a_v(z)u(t)$ and $u(0) = 0$. The formula is entire, including at $a_v(z) = 0$, and

$$q(\Phi_v(z, t)) - 1 = e^{ta_v(z)}(q(z) - 1).$$

Thus every flow preserves U . The composition

$$s(z; t_1, \dots, t_n) = \Phi_{v_n}(\dots \Phi_{v_1}(z, t_1) \dots, t_n)$$

is a holomorphic spray $U \times \mathbb{C}^n \rightarrow U$. Its parameter differential at $(z, 0)$ has columns $(q(z) - 1)v_j$, which form a basis of $T_z U$. The spray is dominating, so U is elliptic and hence Oka by Gromov's Oka principle [5].

Finally,

$$\mathcal{O}_{\mathbb{P}^n}(-(K_{\mathbb{P}^n} + Q + H)) \cong \mathcal{O}_{\mathbb{P}^n}(n - 2),$$

which is ample exactly when $n \geq 3$. Transversality makes $Q + H$ a reduced simple normal crossings divisor, and $[Q] = 2[H]$ in $N^1(\mathbb{P}^n)_{\mathbb{Q}}$. $\square$

7.3. Toric boundary complements.

Proposition 7.4. *Let X be a smooth complex toric variety and let D be a reduced union of torus-invariant prime divisors. Then $X \setminus D$ is Oka. This applies in particular to every reduced torus-invariant log Fano pair on a smooth projective toric variety, with no ampleness or numerical independence assumption on the individual boundary components.*

Proof. The complement is a smooth torus-invariant open subvariety containing the dense torus, hence is itself a smooth toric variety, and is Oka by Lárusson's theorem [19, Theorem 3]. $\square$

ACKNOWLEDGMENTS

We are grateful to Yunheng Du for many discussions over a long period and for generously sharing numerous manuscripts, which have been very helpful in this work. In recognition of his contributions, we invited him to join us as a coauthor, but he graciously declined.

STATEMENT ON THE USE OF ARTIFICIAL INTELLIGENCE

This manuscript grew out of a substantial body of earlier drafts and working notes, discussions within our research group, and sustained interaction with generative artificial intelligence in exploring, organizing, and revising the mathematical arguments and their exposition. The authors have reviewed the resulting text and take full responsibility for its mathematical content and final form.

REFERENCES

- [1] Y.-H. Du, B. Guo, P.-C. Wang, and S.-Y. Xie. *A Strong Dominability Criterion and an Oka Union Theorem*. Unpublished manuscript, October 8, 2026.
- [2] S. An and S.-Y. Xie. *Every smooth complex Fano threefold is Oka*. Unpublished manuscript, October 6, 2026. https://xiesongyan.github.io/papers/an_xie_every_smooth_complex_fano_threefold_oka_2026.pdf.
- [3] S. An and S.-Y. Xie. *Oka approximation on rationally connected threefolds*. Unpublished manuscript. <https://xiesongyan.github.io/>.
- [4] P.-C. Wang, S.-Y. Xie, and S. Zhao. *Oka properties of genus-one K3 surfaces and generalized Kummer surfaces*. Preprint, October 6, 2026. https://xiesongyan.github.io/papers/wang_xie_zhao_oka_genus_one_k3_kummer_2026.pdf.

- [5] M. Gromov. Oka's principle for holomorphic sections of elliptic bundles. *Journal of the American Mathematical Society* **2** (1989), 851–897. doi:10.1090/S0894-0347-1989-1001851-9.
- [6] F. Campana and J. Winkelmann, with an appendix by J. Kollár. Dense entire curves in rationally connected manifolds. *Algebraic Geometry* **10** (2023), 521–553. doi:10.14231/AG-2023-018.
- [7] A. Alarcón and F. Forstnerič. Oka-1 manifolds. *Mathematische Zeitschrift* **311** (2025), article 33. doi:10.1007/s00209-025-03833-4.
- [8] Y.-H. Du, B. Guo, and S.-Y. Xie. *Analytic and Algebraic Oka-1 Approximation for Smooth Projective Morphisms with Rationally Connected Fibers*. arXiv:2609.11883v1, September 10, 2026. <https://arxiv.org/abs/2609.11883v1>.
- [9] F. Forstnerič. Runge approximation on convex sets implies the Oka property. *Annals of Mathematics* **163** (2006), 689–707. doi:10.4007/annals.2006.163.689.
- [10] F. Forstnerič. *Stein Manifolds and Holomorphic Mappings: The Homotopy Principle in Complex Analysis*. Second edition. Springer, 2017. doi:10.1007/978-3-319-61058-0.
- [11] O. Debarre. Variétés rationnellement connexes (d'après T. Graber, J. Harris, J. Starr et A. J. de Jong). *Astérisque* **290** (2003), Séminaire Bourbaki, Exp. No. 905, 243–266. https://www.numdam.org/item/SB_2001-2002__44__243_0/.
- [12] O. Debarre. *Rational curves on algebraic varieties*. Lecture notes for GAeL XVIII and the Strasbourg Semaine spéciale, version November 14, 2020. <https://perso.imj-prg.fr/wp-content/uploads/debarre-pub/NotesGAEL.pdf>.
- [13] J. Kollár. Rationally connected varieties over local fields. *Annals of Mathematics* **150** (1999), 357–367. doi:10.2307/121107.
- [14] F. Campana. Connexité rationnelle des variétés de Fano. *Annales scientifiques de l'École Normale Supérieure* (4) **25** (1992), 539–545. doi:10.24033/asens.1658.
- [15] J. Kollár, Y. Miyaoka, and S. Mori. Rational connectedness and boundedness of Fano manifolds. *Journal of Differential Geometry* **36** (1992), 765–779. doi:10.4310/jdg/1214453188.
- [16] Q. Chen and Y. Zhu. $\mathbb{A}^1$ -curves on log smooth varieties. *Journal für die reine und angewandte Mathematik* **756** (2019), 1–35. doi:10.1515/crelle-2017-0028. https://math-zhu.github.io/papers/CZ_A1_pair.pdf.
- [17] S.-Y. Xie. *Elliptic complements of cubic hypersurfaces*. arXiv:2608.02413v3, revised September 30, 2026, 28 pages. <https://arxiv.org/abs/2608.02413v3>.
- [18] H. Maeda. Classification of logarithmic Fano threefolds. *Compositio Mathematica* **57** (1986), 81–125. https://www.numdam.org/item/CM_1986__57_1_81_0.pdf.
- [19] F. Lárusson. *Smooth toric varieties are Oka*. arXiv:1107.3604v3 (2013). <https://arxiv.org/abs/1107.3604v3>.
- [20] C. Schoen. On fiber products of rational elliptic surfaces with section. *Mathematische Zeitschrift* **197** (1988), 177–199. doi:10.1007/BF01215188.
- [21] R. Donagi, B. A. Ovrut, T. Pantev, and D. Waldram. Spectral involutions on rational elliptic surfaces. *Advances in Theoretical and Mathematical Physics* **5** (2001), 499–561. doi:10.4310/ATMP.2001.v5.n3.a4.
- [22] S.-Y. Xie and S. Zhao. *Periods of Oka K3 surfaces are dense*. arXiv:2608.24392v2, revised September 17, 2026. <https://arxiv.org/abs/2608.24392v2>.
- [23] C. Gachet, H.-Y. Lin, and L. Wang. Nef cones of fiber products and an application to the cone conjecture. *Forum of Mathematics, Sigma* **12** (2024), e28. doi:10.1017/fms.2024.22.
- [24] A. Alzati, E. Ballico, and G. Ottaviani. The theorem of Mather on generic projections for singular varieties. *Geometriae Dedicata* **85** (2001), 113–117. doi:10.1023/A:1010383721612. <https://people.dimai.unifi.it/ottaviani/public/singmather.pdf>.
- [25] J. Carlson and P. Griffiths. A defect relation for equidimensional holomorphic mappings between algebraic varieties. *Annals of Mathematics* (2) **95** (1972), 557–584. doi:10.2307/1970871. <https://publications.ias.edu/sites/default/files/defectrelation.pdf>.

(Sicheng An) SCHOOL OF MATHEMATICAL SCIENCES, UNIVERSITY OF CHINESE ACADEMY OF SCIENCES, BEIJING 100049, CHINA

Email address: `ansicheng23@mailsucas.ac.cn`

(Bin Guo) ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING 100190, CHINA

Email address: `guobin@amss.ac.cn`

(Peng-Chao Wang) ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING 100190, CHINA

Email address: `wangpengchao26@mailsucas.ac.cn`

(Song-Yan Xie) STATE KEY LABORATORY OF MATHEMATICAL SCIENCES, ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING 100190, CHINA; SCHOOL OF MATHEMATICAL SCIENCES, UNIVERSITY OF CHINESE ACADEMY OF SCIENCES, BEIJING 100049, CHINA

Email address: `xiesongyan@amss.ac.cn`